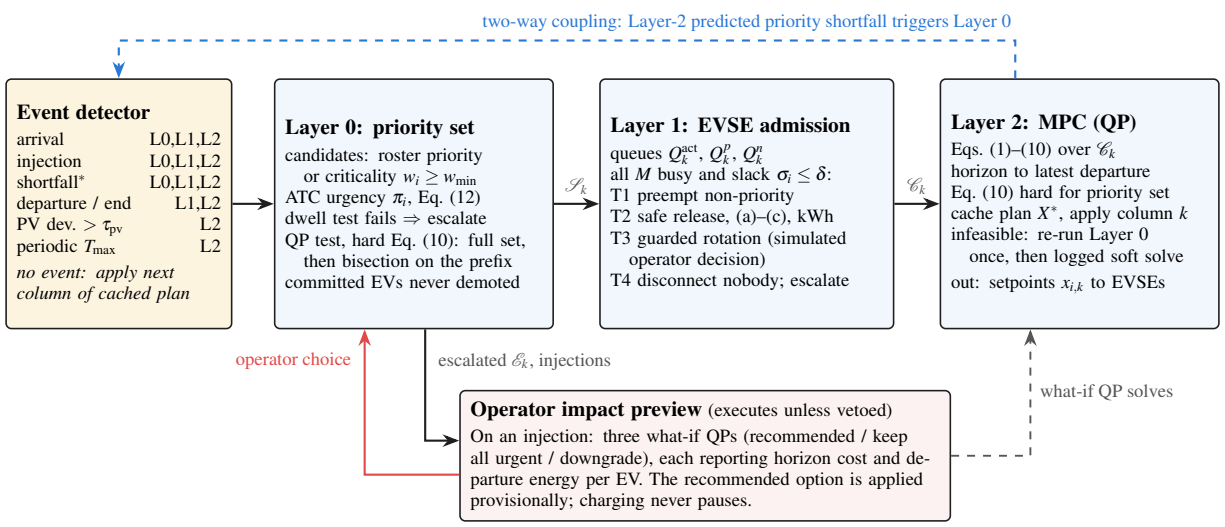

two-way coupling: Layer-2 predicted priority shortfall triggers Layer 0
Event detector
arrival L0,L1,L2
injection L0,L1,L2
shortfall* L0,L1,L2
departure / end L1,L2
PV dev. > τ_{pv} L2
periodic T_{max} L2
no event: apply next column of cached plan
Layer 0: priority set
candidates: roster priority or criticality $w_i \geq w_{min}$
ATC urgency π_i, Eq. (12)
dwell test fails ⇒ escalate
QP test, hard Eq. (10): full set, then bisection on the prefix
committed EVs never demoted
$\mathscr{S}_k$
Layer 1: EVSE admission
queues Q_k^{act}, Q_k^p, Q_k^n
all M busy and slack $\sigma_i \leq \delta$:
T1 preempt non-priority
T2 safe release, (a)–(c), kWh
T3 guarded rotation (simulated operator decision)
T4 disconnect nobody; escalate
$\mathscr{C}_k$
Layer 2: MPC (QP)
Eqs. (1)–(10) over $\mathscr{C}_k$
horizon to latest departure
Eq. (10) hard for priority set
cache plan X^*, apply column k
infeasible: re-run Layer 0 once, then logged soft solve
out: setpoints $x_{i,k}$ to EVSEs
operator choice
escalated $\mathscr{C}_k$, injections
what-if QP solves
Operator impact preview (executes unless vetoed)
On an injection: three what-if QPs (recommended / keep all urgent / downgrade), each reporting horizon cost and departure energy per EV. The recommended option is applied provisionally; charging never pauses.